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CHAPTER 1 • NUMBER • TOPIC 3

Prime Factors, HCF & LCM

Break every number into primes. Then HCF takes the lowest powers of what they share, LCM takes the highest powers of everything.



$$\frac{3}{10} = 0.3 =$$

30%

5

VARIANTS

29

PAST-PAPER PARTS

37

PRACTICE QUESTIONS

YOU WILL MASTER

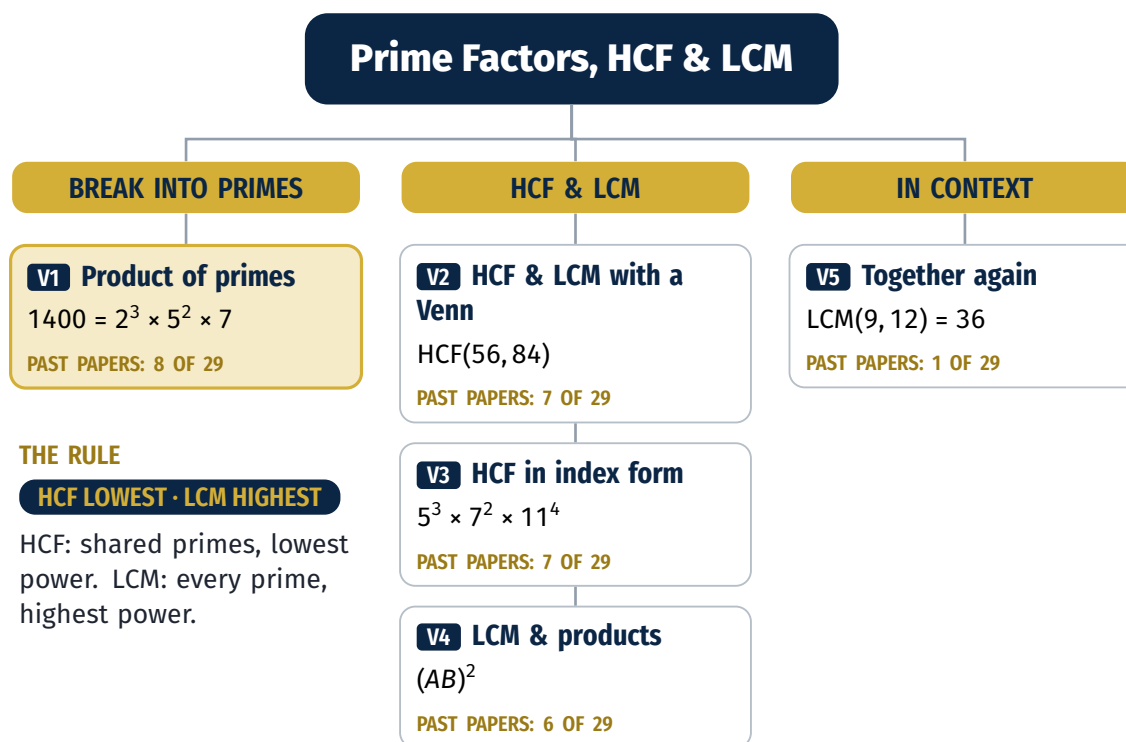
V1 Product of primes V2 HCF & LCM: Venn V3 HCF in index form

V4 LCM & products in index form V5 Together again

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Prime Factors, HCF & LCM at a glance



Your notes


★ How often does each variant come up?

22 past-paper questions (Jan 2020 → Jun 2025) hold 29 parts; 37 practice parts — grouped by type.

	PAST PAPERS (OUT OF 29)	PRACTICE PARTS
V1 Product of primes	8	10
V2 HCF & LCM with a Venn	7	11
V3 HCF in index form	7	4
V4 LCM & products	6	4
V5 Together again	1	12

WHAT THIS TELLS YOU This topic is Question 1 or 2 of many papers: “write 1400 as a product of powers of its prime factors” (3 marks). Index-form HCF/LCM appears almost every year too.

BEFORE YOU START

The big idea



Your notes
Every number is built from primes.

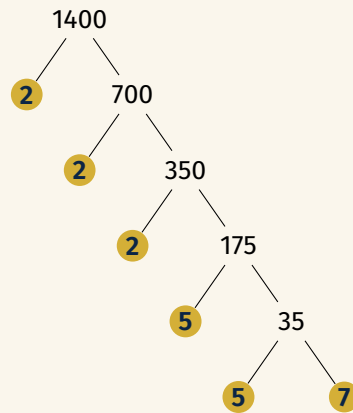
Like a recipe:

$$1400 = 2 \times 2 \times 2 \times 5 \times 5 \times 7 = 2^3 \times 5^2 \times 7.$$

There is only **one** recipe for each number.

Primes to know:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29.

**HCF and LCM****HCF — Highest Common Factor**shared primes, **lowest** powerThe biggest number that divides into both.
Always $\leq$ the smaller number.**LCM — Lowest Common Multiple**every prime, **highest** powerThe smallest number both divide into.
Always $\geq$ the bigger number.**Index laws you will need****Multiply**

$$5^{17} \times 5^2 = 5^{19}$$

add the powers

Power of a product

$$(2^3 \times 5)^2 = 2^6 \times 5^2$$

multiply every power

Hidden primes

$$16 = 2^4$$

rewrite first

**CALCULATOR FIRST, THEN CHECK**

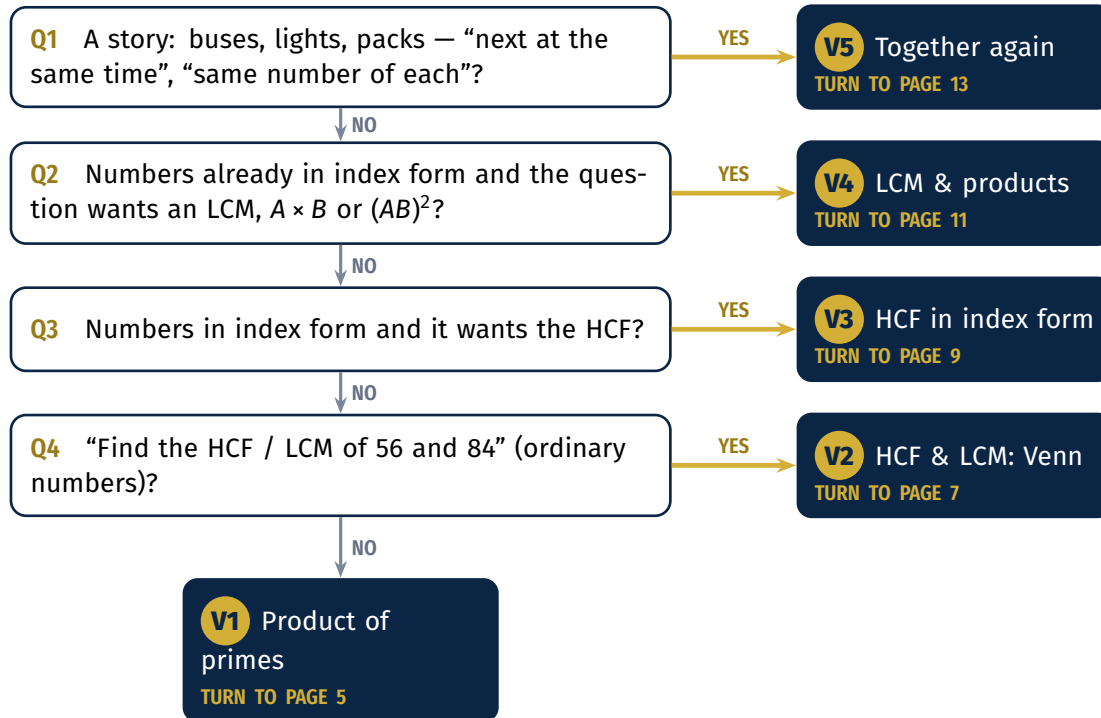
Calculator first: type the number, [=], then **SHIFT** **FACT** — it shows $2^3 \times 5^2 \times 7$. Then write the factor tree: “show your working clearly” means the tree (or the ladder) must be on the page. Check an HCF/LCM by multiplying out.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1 MOST COMMON IN THE EXAM

Product of Prime Factors

Factor tree (or ladder), then write it with powers.

SEEN IN PAST PAPERS

8 / 29

PRACTICE QUESTIONS: 10

 Your notes

Came up in: Jan 2020 P2HR Q1(b) • May 2021 P2H Q1 • Nov 2021 P1H Q7 • Jun 2022 P2HR Q2 • May 2023 P1HR Q1 • May 2023 P1H Q8(a) • Jun 2024 P2HR Q1 • Jun 2025 P2HR Q2

? What is it?

In plain words

Split the number into two factors again and again until every branch ends in a **prime** (circle it).

Collect the circled primes in order and write repeats as **powers**:

$$2 \times 2 \times 2 = 2^3.$$

IT LOOKS LIKE

800

1400

2250

 3.6×10^3

2	1400
2	700
2	350
5	175
5	35
7	7

DIVIDE BY PRIMES,
SMALLEST FIRST

1

The ladder: divide by the smallest prime you can.

How do I do it?

1 Split

Any two factors: $1400 = 2 \times 700$ (or 14×100).

2 Keep splitting

Until every branch ends in a prime. Circle the primes.

3 Collect

List the primes smallest first.

4 Powers

Repeats as powers: $2^3 \times 5^2 \times 7$. Check: multiply back.

CIRCLE THE PRIMES

"Product of powers" means use indices.

V1 Product of Prime Factors

VARIANT 1 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Write 1400 as a product of powers of its prime factors” → **Tree, then $2^3 \times 5^2 \times 7$**

“Show your working clearly” → **The tree or ladder must be shown**

“ 3.6×10^3 ” → **Write it as 3600 first**

WORKED EXAMPLE PAST PAPER

June 2024 • Paper 2HR • Q1 • 3 marks

Write 1400 as a product of powers of its prime factors. Show your working clearly.

- STEP 1** $1400 = 2 \times 700$ ◀ split
- STEP 2** $700 = 2 \times 350$, $350 = 2 \times 175$, $175 = 5 \times 35$, $35 = 5 \times 7$ ◀ keep splitting to primes
- STEP 3** $2 \times 2 \times 2 \times 5 \times 5 \times 7$ ◀ collect, smallest first
- STEP 4** $2^3 \times 5^2 \times 7$ ◀ powers

ANSWER $2^3 \times 5^2 \times 7$ ✓ $8 \times 25 \times 7 = 1400$ ✓**WATCH OUT**

× $2^3 \times 5^2 \times 7 = 2 \times 2 \times 2 \times 5 \times 5 \times 7$ only

✓ write it with powers

“Product of **powers**” — the index form is the final answer.

EXAM TIP

8 of the 29 past-paper parts — 3 easy marks. Check by multiplying back on the calculator.

Check yourself

calculator first • answers on page 15

1.1 May 2021 P2H

Write 600 as a product of powers of its prime factors.

Answer:

1.2 May 2023 P1HR

Write 2250 as a product of powers of its prime factors.

Answer:

1.3 Nov 2021 P1H

Write 3.6×10^3 as a product of powers of its prime factors.

Answer:

V2

VARIANT 2

HCF & LCM with a Venn Diagram

Put the prime factors in a Venn: the middle is the HCF, everything is the LCM.

SEEN IN PAST PAPERS

7 / 29

PRACTICE QUESTIONS: 11



Your notes

Came up in: Jan 2020 P1H Q2 • Jan 2022 P2H Q7 • Jan 2022 P1HR Q6(a) • Jun 2022 P1H Q3(a)(b) • May 2024 P1H Q3 • Jun 2025 P1HR Q1

? What is it?**In plain words**

Write both numbers as primes.
Shared primes go in the **middle**,
the rest on each side.

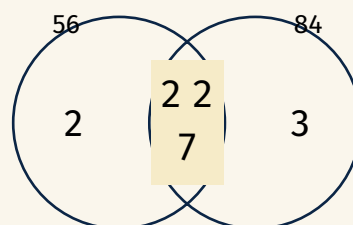
HCF = multiply the middle. **LCM** =
multiply **everything**.

IT LOOKS LIKE

HCF(56, 84)

LCM(60, 72)

LCM(28, 42, 63)



$$\text{HCF} = 2 \times 2 \times 7 = 28$$

$$\text{LCM} = 2 \times 2 \times 2 \times 7 \times 3 = 168$$

How do I do it?**1 Primes**

Write each number as a product of primes (V1).

2 Venn

Shared primes in the middle (once each pair), the rest outside.

3 HCF

Multiply the middle.

4 LCM

Multiply everything in the diagram.

HCF = MIDDLE • LCM = EVERYTHINGCheck: $\text{HCF} \times \text{LCM} = \text{the two numbers multiplied.}$

V2 HCF & LCM with a Venn Diagram

VARIANT 2 • SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

"Find the HCF of 56 and 84"

→ Venn: multiply the middle

"Find the LCM of 60 and 72"

→ Venn: multiply everything

Three numbers: LCM of 28, 42, 63

→ Highest power of every prime

**WORKED EXAMPLE****PAST PAPER**

June 2022 • Paper 1H • Q3 • 4 marks

(a) Find the HCF of 56 and 84. (b) Find the LCM of 60 and 72. Show your working clearly.

STEP 1 $56 = 2^3 \times 7$, $84 = 2^2 \times 3 \times 7$ ◀ (a) primes**STEP 3** $HCF = 2^2 \times 7 = 28$ ◀ (a) shared, lowest power**STEP 1** $60 = 2^2 \times 3 \times 5$, $72 = 2^3 \times 3^2$ ◀ (b) primes**STEP 4** $LCM = 2^3 \times 3^2 \times 5 = 360$ ◀ (b) every prime, highest power**ANSWER** (a) 28 (b) 360✓ $56 \div 28 = 2$, $360 \div 60 = 6$, $360 \div 72 = 5$ **WATCH OUT**✗ $LCM(60, 72) = 60 \times 72 = 4320$

✓ 360

Multiplying gives a common multiple, not the **lowest**.**EXAM TIP**

Check: the HCF divides both numbers; both numbers divide the LCM.

**Check yourself**

calculator first • answers on page 15

2.1

May 2024 P1H

Find the HCF of 72 and 108.

Answer:

2.2

Jan 2020 P1H

Find the LCM of 28 and 105.

Answer:

2.3

Jan 2022 P2H

Find the LCM of 28, 42 and 63.

Answer:

V3

VARIANT 3

HCF in Index Form

Numbers given as powers of primes: take only the shared primes, each at its lowest power.

SEEN IN PAST PAPERS

7 / 29

PRACTICE QUESTIONS: 4



Your notes

Came up in: Oct/Nov 2020 P1H Q10(a) • Jan 2021 P1HR Q9(a) • Jan 2022 P1HR Q6(b) • Jun 2022 P1HR Q12(a) • Nov 2023 P1H Q7 • May 2024 P1HR Q11(a) • Nov 2024 P1H Q6

? What is it?**In plain words**

Line the primes up in a table. For each prime that is in **every** number, take the **smallest power**. A prime missing from one number is left out.

Leave the answer as powers — do not multiply out.

IT LOOKS LIKE

$$A = 5^3 \times 7^3 \times 11^6$$

$$B = 5^6 \times 7^2 \times 11^4$$

	prime 5	prime 7	prime 11
A	5^3	7^3	11^6
B	5^6	7^2	11^4
HCF	5^3	7^2	11^4

LOWEST POWER IN EACH COLUMN

How do I do it?**1 Rewrite**

Make every factor a prime power: $16 = 2^4$, 5A adds a 5.

2 Table

One column per prime.

3 Lowest power

Only primes in every number; take the smallest index.

4 Answer

Multiply the chosen prime powers (leave in index form).

HCF = SHARED, LOWEST

A prime in only one number is not common.

V3 HCF in Index Form

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Find the HCF of A and B” (given as powers)

→ **Shared primes, lowest power**

“HCF of $5A$ and $2B$ ”

→ **Multiply in the 5 and the 2 first**

$B = 16 \times 3^{37}$

→ **Rewrite:** $2^4 \times 3^{37}$

WORKED EXAMPLE **PAST PAPER**

November 2023 • Paper 1H • Q7 • 2 marks

$A = 5^3 \times 7^3 \times 11^6$ and $B = 5^6 \times 7^2 \times 11^4$. Find the HCF of A and B. Give your answer as a product of powers of its prime factors.

STEP 2 $5 : 3 \text{ vs } 6, 7 : 3 \text{ vs } 2, 11 : 6 \text{ vs } 4$ ◀ table of powers

STEP 3 $5^3, 7^2, 11^4$ ◀ lowest power of each

STEP 4 $5^3 \times 7^2 \times 11^4$ ◀ multiply the chosen powers

ANSWER $5^3 \times 7^2 \times 11^4$

✓ it divides both A and B

WATCH OUT

✗ $\text{HCF} = 5^6 \times 7^3 \times 11^6$

✓ **lowest powers:** $5^3 \times 7^2 \times 11^4$

Highest powers make the LCM, not the HCF.

EXAM TIP

A factor must fit **inside** both numbers — so its powers can't be bigger than either.

Check yourself

calculator first • answers on page 15

3.1 Nov 2024 P1H

$A = 2^3 \times 5^4 \times 7 \times 11,$
 $B = 2^2 \times 5^2 \times 7^2,$
 $C = 2^2 \times 5^3 \times 7^4.$ HCF of A, B, C?

Answer:

3.2 Jun 2022 P1HR

$P = 3^3 \times 5^2 \times 7,$
 $Q = 3^2 \times 5 \times 7^2.$ Write down the HCF of P and Q.

Answer:

3.3 Oct/Nov 2020 P1H

$A = 2 \times 3^{43}, B = 16 \times 3^{37}.$
 Find the HCF of A and B.

Answer:

V4

VARIANT 4

LCM & Products in Index Form

LCM: every prime at its highest power.

Products: add the powers.

SEEN IN PAST PAPERS

6 / 29

PRACTICE QUESTIONS: 4

 Your notes

Came up in: Oct/Nov 2020 P1H Q10(b) • Jan 2021 P1HR Q9(b) • Jan 2021 P2H Q9 • Jun 2022 P1HR Q12(b) • May 2023 P1H Q8(b) • May 2024 P1HR Q11(b)

? What is it?

In plain words

LCM: take **every** prime that appears anywhere, at its **highest** power.

Products ($A \times B$, $P^3 \times Q$, $(AB)^2$): use the index laws — **add** powers when multiplying, **multiply** powers for a power.

IT LOOKS LIKE

LCM(A , B) $A \times B$ $P^3 \times Q$ $(AB)^2$

	2	3	5	11	13
A	2^3	3^2	5^2	11	–
B	2^4	3	5^4	–	13
LCM	2^4	3^2	5^4	11	13

HIGHEST POWER, EVERY PRIME

How do I do it?

1 Rewrite

Bring in extra factors: $2A$, $3B$, $5A$, $7B$.

2 LCM

Every prime, highest index (a prime in only one number still goes in).

3 Products

$\times$: add the powers of each prime. $()^2$: double every power.

4 Answer

As a product of prime powers, primes in order.

LCM = EVERY PRIME, HIGHEST

Products: index laws.

V4 LCM & Products in Index Form

VARIANT 4 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Find the LCM of A and B ... as a product of powers”

→ **Every prime, highest power**

“Express $A \times B$...” / “ $P^3 \times Q$ ”

→ **Add the powers of each prime**

“(AB)²”

→ **Find AB, then double every power**

**WORKED EXAMPLE** **PAST PAPER**

January 2021 • Paper 2H • Q9 • 2 marks

$A = 2^3 \times 3^2 \times 5^2 \times 11$, $B = 2^4 \times 3 \times 5^4 \times 13$. Find the LCM of A and B as a product of powers of prime numbers.

STEP 2

2, 3, 5, 11, 13

◀ every prime that appears

STEP 3 2^4 , 3^2 , 5^4 , 11, 13

◀ highest power of each

STEP 4 $2^4 \times 3^2 \times 5^4 \times 11 \times 13$

◀ multiply

ANSWER $2^4 \times 3^2 \times 5^4 \times 11 \times 13$

✓ both A and B divide it

**WATCH OUT**

✗ leaving out 11 and 13

✓ include every prime

11 is only in A and 13 only in B — the LCM must contain both numbers.

**EXAM TIP**

“HCF of 5A and 2B”: multiply in the 5 and 2 **first**, then compare powers.

**Check yourself**

calculator first • answers on page 15

4.1

May 2024 P1HR

$A = 2^5 \times 5 \times 7^2$,
 $B = 2^3 \times 5^3 \times 7^4$. Find
 $(AB)^2$ as a product of
 prime factors.

Answer:

4.2

Jun 2022 P1HR

$P = 3^3 \times 5^2 \times 7$,
 $Q = 3^2 \times 5 \times 7^2$. Find
 $P^3 \times Q$.

Answer:

4.3

Oct/Nov 2020 P1H

$A = 2 \times 3^{43}$, $B = 16 \times 3^{37}$.
 Express $A \times B$ as a
 product of powers.

Answer:

V5

VARIANT 5

Together Again: LCM Problems

“When do they next happen at the same time?” or “the same number of each” — it is an LCM.

SEEN IN PAST PAPERS

1 / 29

PRACTICE QUESTIONS: 12

 Your notes

Came up in: Jun 2023 P2HR Q2

? What is it?

In plain words

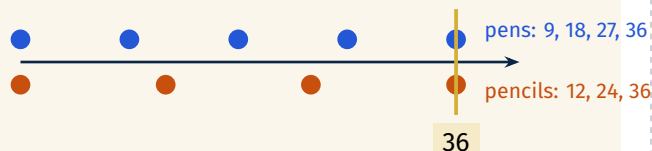
Things that repeat — buses every 12 and 18 minutes, packs of 9 and 12 — meet again at the **LCM**.

Then finish the story: the time, the number of packs, the cost.

IT LOOKS LIKE

buses every 12 and 18 min

packs of 9 and 12



How do I do it?

1 Spot it

“Same time again”, “same number of each”, “together” → LCM.

2 LCM

Venn or list the multiples.

3 How many of each

$\text{LCM} \div \text{each group size}$.

4 Finish

Time, cost or count — what the question asks.

TOGETHER AGAIN = LCM Sharing equally into groups = HCF.

V5 Together Again: LCM Problems

VARIANT 5 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO*"next leave at the same time"*→ **LCM of the two times, then add to the start***"exactly the same number of pens as pencils"*→ **LCM of the pack sizes***"the largest number of identical groups"*→ **HCF instead****WORKED EXAMPLE** **PAST PAPER**

June 2023 • Paper 2HR • Q2 • 4 marks

Packets of 9 pens cost £7.60; boxes of 12 pencils cost £4.80. Sandeep buys full packets and boxes and wants exactly the same number of pens as pencils. Find the minimum amount he pays.

STEP 2 $\text{LCM}(9, 12) = 36$

◀ same number of each → LCM

STEP 3 $36 \div 9 = 4$ packets, $36 \div 12 = 3$ boxes

◀ how many of each

STEP 4 $4 \times 7.60 + 3 \times 4.80 = 30.40 + 14.40$

◀ the cost

ANSWER **£44.80**

✓ 36 pens and 36 pencils ✓

WATCH OUT $\times 9 \times 12 = 108$ pens✓ **LCM = 36**

The question says **minimum** — use the lowest common multiple.

EXAM TIP

Write the LCM line first — it shows the examiner the method even if the money slips.

Check yourself

calculator first • answers on page 15

5.1

Dr Eslam

Bus A leaves every 12 min, bus B every 18 min. Both leave at 9:00 am. When next together?

Answer:

5.2

Dr Eslam

Lights flash every 15 s and every 20 s. Together at 9:00:00. When next together?

Answer:

5.3

Dr Eslam

Matt runs a lap in 50 s, Dan in 80 s, starting together. Laps each when they next meet at the start?

Answer:

CHECK YOURSELF

Answers



Every answer below was checked with sympy (check_answers.py).

1.1 $2^3 \times 3 \times 5^2$

1.2 $2 \times 3^2 \times 5^3$

1.3 $2^4 \times 3^2 \times 5^2$

2.1 36

2.2 420

2.3 252

3.1 $2^2 \times 5^2 \times 7$

3.2 $3^2 \times 5 \times 7$

3.3 2×3^{37}

4.1 $2^{16} \times 5^8 \times 7^{12}$

4.2 $3^{11} \times 5^7 \times 7^5$

4.3 $2^5 \times 3^{80}$

5.1 9:36 am

5.2 9:01:00

5.3 Matt 8, Dan 5

THE WHOLE TOPIC ON ONE PAGE

Prime Factors, HCF & LCM cheat sheet



Your notes
V1 Product of primes 8/29

$$1400 = 2^3 \times 5^2 \times 7$$

Tree or ladder; circle primes; powers.

V2 Venn 7/29

$$\text{HCF}(56, 84) = 28$$

Middle = HCF, everything = LCM.

V3 HCF index form 7/29

$$5^3 \times 7^2 \times 11^4$$

Shared primes, lowest power.

V4 LCM & products 6/29

$$2^4 \times 3^2 \times 5^4 \times 11 \times 13$$

Every prime, highest power; $\times$ adds powers.**V5 Together again 1/29**

$$\text{LCM}(9, 12) = 36$$

Same time / same number $\rightarrow$ LCM.**TOP 3 LOST MARKS**

1. no powers in the final answer $\times$
2. HCF and LCM swapped $\times$
3. LCM = just multiply $\times$

BOTTOM LINE Primes first • HCF: shared, lowest power • LCM: everything, highest power • together again = LCM.